WHAT IS A REFERENCE ARCHITECTURE?

A reference architecture is a standardized, high-level design blueprint that outlines the essential components, roles, and relationships in a specific type of system. It does not prescribe a single implementation but instead provides a template or best-practice guide for structuring systems to address a particular class of problems. Reference architectures possess several important properties:

Abstract yet actionable: It identifies the necessary components (for example, Policy Enforcement Point, Policy Decision Point) and explains their interactions, illustrating their importance without dictating specific tools or code.

Reusable across domains: A reference architecture can be used in various contexts (for example, cloud platforms, internal apps, APIs), offering a consistent language for teams to design, build, and evaluate systems.

Guides implementation: It assists architects, engineers, and decision-makers in aligning on how to break down responsibilities, reduce duplication, and enforce consistent security or functional patterns.

Reference architectures provide a solid mental model that helps us discuss, analyze, and understand IT systems.

Figure 4.2 illustrates the authorization reference architecture that we will use to discuss authorization systems throughout the book. The terminology in this reference architecture is widely used, and references to it can be found in system documentation, blog posts, and even sales literature. Nowadays, we can see this clear separation of enforcement, decision-making, policy management, and attribute retrieval in systems like AWS Verified Permissions, Open Policy Agent (OPA), and Aserto’s Topaz.